

Rainfall Anomalies and ENSO in Vietnam

An Analysis of 2025 Rainfall Conditions and Historical ENSO Relationships

INDEPENDENT RESEARCH PROJECT

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ABSTRACT

This independent research project examines how unusual rainfall conditions were in Vietnam in 2025 and the extent to which they were associated with the historical El Niño–Southern Oscillation (ENSO) signal. Monthly precipitation data from ERA5-Land and Niño 3.4 sea surface temperature data from NOAA ERSSTv5 are used to analyse rainfall anomalies, the Standardized Precipitation Index (SPI), regional and spatial ENSO relationships, and lagged associations. The results show that the strongest positive rainfall anomalies in 2025 were concentrated in central Vietnam, particularly in North Central and South Central Vietnam. Historically, the ENSO–rainfall relationship is weak in northern Vietnam, stronger across central and southern Vietnam, and most apparent at three- to six-month precipitation accumulation periods. When 2025 conditions are compared with the historical ENSO–SPI relationship, North Central Vietnam is significantly wetter than the historical ENSO-based benchmark at both SPI-3 and SPI-6, while South Central Vietnam shows a significant departure at SPI-3. Local case studies further show that ENSO provides only limited information about individual monthly rainfall extremes. Overall, ENSO is useful for understanding broad seasonal and regional rainfall variability, but it does not fully account for the unusually wet conditions observed in central Vietnam in 2025.

1. Introduction

Vietnam experiences substantial rainfall variability across regions and seasons. Periods of unusually high rainfall can increase flood risk, disrupt transport and infrastructure, affect households and economic activity, and create wider social and financial consequences. Understanding rainfall variability is therefore relevant not only for climate science, but also for climate-risk assessment and adaptation planning.

During 2025, several parts of Vietnam experienced episodes of heavy rainfall and flooding, raising the question of whether observed rainfall conditions were unusual relative to the historical climate. The impacts were particularly visible in central Vietnam, including Hue and Khanh Hoa, making the region an important focus for examining both the magnitude of rainfall anomalies and the broader climate conditions associated with them.

One large-scale climate pattern that may help explain part of this variability is the El Niño–Southern Oscillation (ENSO). ENSO is associated with changes in precipitation across many parts of the world, although the strength and direction of this relationship vary across regions and seasons (WMO, 2024). Its influence on rainfall in Vietnam is therefore unlikely to be uniform across the country or across different time scales.

This study addresses two main questions:

- 1. How unusual were rainfall conditions in Vietnam in 2025 relative to the historical climate?**
- 2. To what extent were the observed rainfall conditions associated with the historical ENSO–rainfall relationship?**

The analysis first establishes the historical rainfall climatology across Vietnam and evaluates annual and spatial rainfall anomalies in 2025. The Standardized Precipitation Index (SPI) is then used to examine wet and dry conditions across different accumulation periods, allowing short-term rainfall variability to be distinguished from more persistent seasonal and multi-month conditions. Historical ENSO–rainfall relationships are subsequently examined across regions, accumulation periods, time lags and individual grid cells.

The 2025 observations are then compared with the historical ENSO–SPI relationship to assess whether rainfall conditions were within the range normally associated with ENSO. Particular attention is given to SPI-3 and SPI-6, which capture seasonal and multi-month rainfall conditions and show the clearest historical relationships with ENSO in several parts of Vietnam.

The regional analysis is complemented by three local case studies. Hue in October 2025 and Khanh Hoa in November 2025 are examined as central Vietnam flood cases, while Ho Chi Minh City in July 2026 provides an additional urban rainfall case outside the main 2025 analysis. In each case, monthly rainfall is compared with its historical climatology and distribution, and contemporaneous and lagged ENSO conditions are evaluated against the historical local ENSO–rainfall relationship.

The objective is to determine where rainfall in 2025 was most unusual, identify the spatial and temporal scales at which ENSO is most closely associated with rainfall in Vietnam, and assess where the observed wet conditions departed from the historical ENSO signal. The case studies then provide a first step toward a broader flood-risk interpretation in which rainfall hazard is considered alongside local flood mechanisms, exposure and vulnerability.

2. Climate Background: ENSO and Rainfall

2.1 El Niño Southern Oscillation

The El Niño Southern Oscillation (ENSO) is a recurring mode of climate variability linked to changes in sea surface temperatures (SSTs) across the tropical Pacific and associated changes in atmospheric pressure and winds. Under normal conditions, easterly trade winds push warm surface water toward the western Pacific, around Indonesia and northern Australia. The warmer ocean in this region supports stronger evaporation, rising moist air and convective rainfall, while colder water is brought to the surface in the eastern Pacific near Peru and Ecuador through upwelling.

During El Niño, the central and eastern equatorial Pacific become warmer than normal and the trade winds weaken. Warm water and atmospheric convection shift eastward, so the western Pacific, including Indonesia and parts of Australia, tends to become drier, while wetter conditions become more likely over the central and eastern Pacific and along parts of the west coast of South America, including Peru and Ecuador. During La Niña, the pattern broadly reverses. The central and eastern Pacific become cooler than normal, upwelling strengthens, and warm water remains more concentrated in the western Pacific. This tends to favour wetter conditions around Indonesia, northern Australia and the western Pacific, while the central and eastern Pacific become relatively drier.

2.2 Measuring ENSO

ENSO conditions are commonly monitored using sea surface temperature anomalies in the equatorial Pacific. One widely used area is the Niño 3.4 region, located approximately between 5°S–5°N and 170°W–120°W. A positive SST anomaly means that sea surface temperatures in the Niño 3.4 region are warmer than the historical average, while a negative SST anomaly means that they are cooler than the historical average. Positive anomalies are associated with the El Niño side of ENSO, while negative anomalies are associated with the La Niña side.

As a general reference, Niño 3.4 SST anomalies around +0.5°C or above are commonly associated with El Niño conditions, while anomalies around –0.5°C or below are associated with La Niña conditions. These temperature thresholds are not applied directly to the ENSO Index used in this study because the Niño 3.4 anomaly is standardized into a z-score. The resulting index is expressed in standard deviation units rather than degrees Celsius. ENSO is therefore represented by a standardized monthly Niño 3.4 SST anomaly derived from NOAA ERSSTv5 data. Positive values indicate relatively warmer Niño 3.4 conditions, negative values indicate relatively cooler conditions, and values close to zero represent conditions near the historical mean.

2.3 ENSO and Rainfall in Vietnam

Vietnam lies within the broader Southeast Asian and western Pacific region affected by changes in tropical Pacific conditions. As ENSO shifts the location of warm water, convection and rainfall across the Pacific, it can also influence rainfall conditions over Vietnam. However, this influence can vary across the country because rainfall in Vietnam differs substantially by region and season, while local conditions are also shaped by monsoon circulation, tropical disturbances, moisture transport and topography.

ENSO therefore provides a large scale climate signal, but it does not determine individual rainfall events. This motivates a regional analysis of the relationship between ENSO and rainfall across Vietnam.

3. Data and Methodology

3.1 Data

Monthly precipitation data are obtained from ERA5-Land and cover Vietnam from January 1950 to August 2026. Total precipitation is converted from metres to millimetres per month using the number of days in each month. The analysis is conducted both at the grid-cell level and for six broad regions: Northern Mountains, Red River Delta, North Central Vietnam, South Central Vietnam, Central Highlands and Southern Vietnam. Regional precipitation is calculated using area-weighted averages to account for differences in grid-cell area across latitude.

ENSO conditions are measured using monthly sea surface temperature data from NOAA ERSSTv5. Sea surface temperatures are averaged over the Niño 3.4 region and converted into monthly anomalies relative to the historical seasonal cycle. The resulting anomaly series is standardized for use in the statistical analysis.

3.2 Rainfall Climatology and Anomalies

Rainfall in Vietnam varies strongly across both space and season. A rainfall total that is unusually high in one region or month may therefore be normal in another. Historical climatologies are constructed separately for each calendar month so that observed rainfall is compared with conditions normally experienced during the same part of the year.

Monthly precipitation anomalies are calculated as:

$$P'_t = P_t - \bar{P}_m$$

where P_t is precipitation in month t , and $\bar{P}_m$ is the historical mean precipitation for the corresponding calendar month.

For the analysis of 2025, historical comparisons use the 1950–2024 period. For observations in 2026, the historical reference period is extended through 2025. Annual precipitation anomalies are also calculated to identify areas with relatively wet or dry conditions across Vietnam.

3.3 Standardized Precipitation Index

Absolute rainfall values are useful for measuring how much water fell, but they are difficult to compare across regions and seasons. An additional 100 mm of rainfall, for example, may represent an unusually wet month in a relatively dry region but may be much less exceptional in a region that normally receives several hundred millimetres of rainfall. Raw precipitation is also highly variable from month to month, making it difficult to distinguish a short-lived rainfall spike from a more persistent wet period.

The Standardized Precipitation Index (SPI) addresses these problems by expressing precipitation relative to its own historical distribution. An important feature of SPI is that precipitation can be accumulated over different time windows before it is standardized. The choice of accumulation period determines the type of wet or dry conditions captured by the index.

SPI-1 is based on precipitation accumulated over one month and therefore responds quickly to short-term rainfall fluctuations. A single unusually wet or dry month can produce a relatively large change in SPI-1. SPI-3 combines rainfall over the previous three months, reducing some of the month-to-month variation and providing a clearer indication of conditions developing over a season. SPI-6 accumulates rainfall over six months and smooths the series further, making it less sensitive to isolated monthly spikes and more representative of persistent wet or dry conditions. Longer accumulation periods, such as SPI-24, capture multi-year precipitation conditions and change more gradually.

Figure 1 illustrates this effect for South Central Vietnam. Monthly precipitation contains frequent and sharp fluctuations, while the three-month accumulated series smooths part of this short-term variation. The six-month series is smoother still because each observation incorporates rainfall from a longer period. The accumulation window is therefore important for interpretation: shorter windows capture rapid rainfall variability, while longer windows emphasize sustained precipitation conditions.

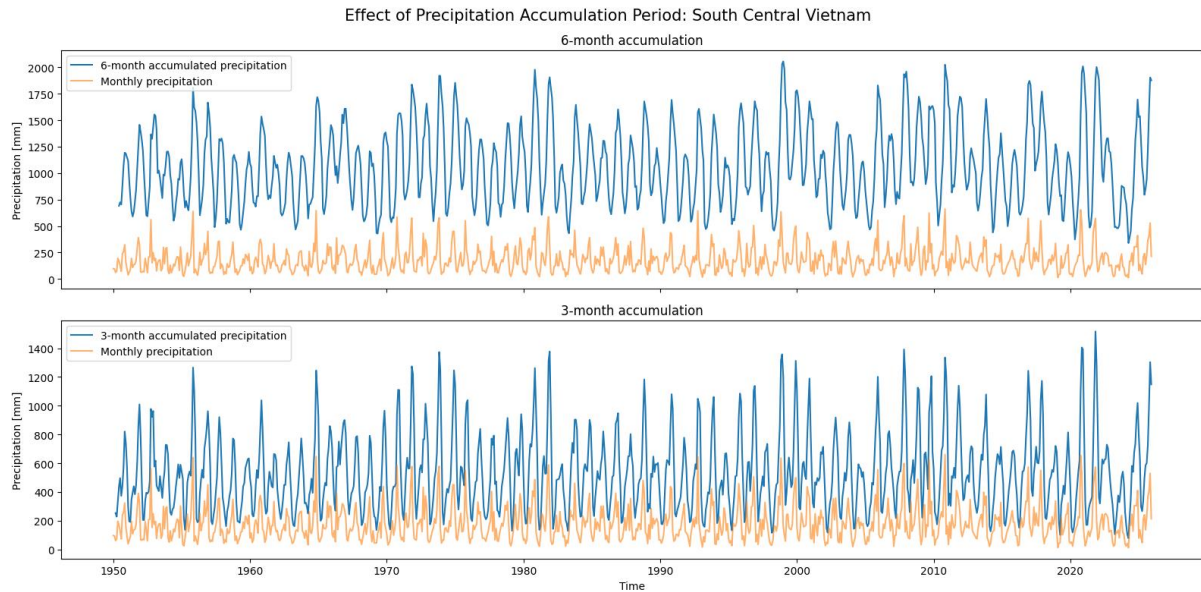


Figure 1: Monthly and Accumulated Precipitation in South Central Vietnam

After precipitation has been accumulated at the selected time scale, it must be converted into a standardized measure. A simple comparison with the mean is not sufficient because precipitation is not normally distributed. Rainfall is bounded at zero and is typically right-skewed because many observations are zero or relatively small, reflecting periods with little or no rainfall, while a much smaller number of observations contain very large rainfall totals. Accumulated precipitation is therefore fitted to a Gamma distribution separately for each calendar month, which is well suited to this type of positive and skewed rainfall distribution.

Figure 2 illustrates the Gamma fit using October precipitation in South Central Vietnam. The fitted Gamma distribution captures the overall shape of the observed rainfall distribution, and the K-S diagnostic gives a p-value of approximately 0.85, so the Gamma fit is not rejected for this example.

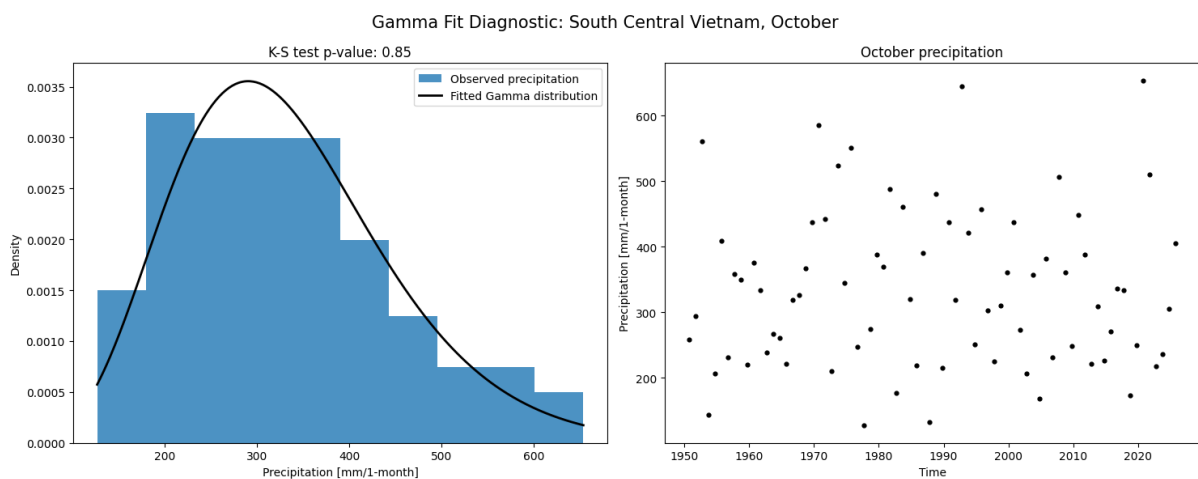


Figure 2. Gamma Fit Diagnostic for October Precipitation in South Central Vietnam

The Gamma-based cumulative probabilities are then mapped onto a standard normal distribution to obtain SPI. **Figure 3** illustrates this transformation from precipitation to probability and finally to standardized SPI values.

The resulting SPI scale is centred around zero. Positive values represent wetter than normal conditions and negative values represent drier than normal conditions. Because rainfall has been expressed relative to its own historical distribution and transformed onto a common standardized scale, SPI values can be compared across regions and accumulation periods even when their underlying rainfall levels differ substantially.

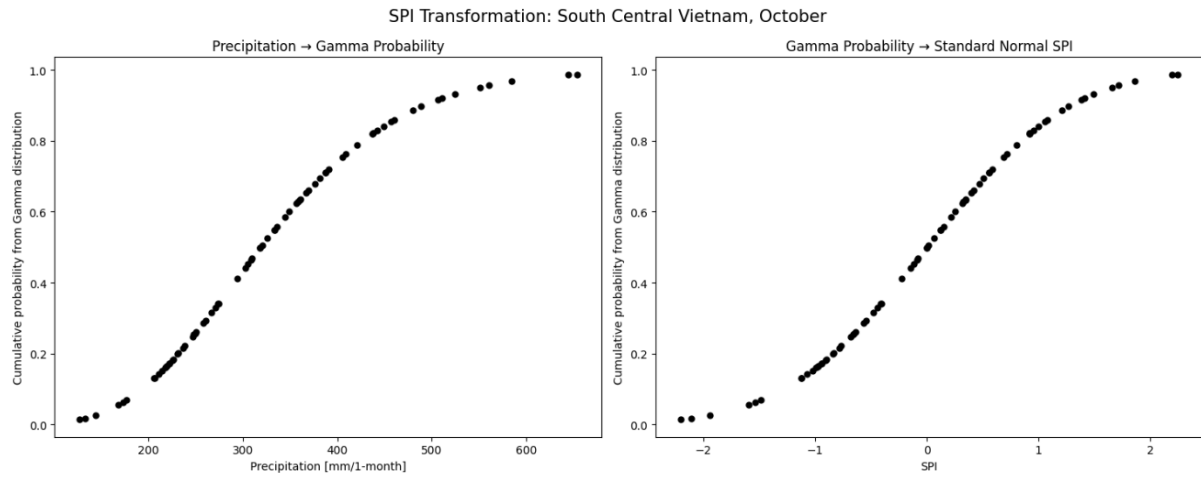


Figure 3. Transformation from Precipitation to SPI in South Central Vietnam

3.4 ENSO Index

ENSO is represented using sea surface temperature anomalies in the Niño 3.4 region. The monthly climatology is first removed from the Niño 3.4 sea surface temperature series to account for the seasonal cycle. The resulting anomaly series is then standardized as:

$$ENSO_t = \frac{SST'_t - \overline{SST'}}{\sigma_{SST'}}$$

where SST'_t is the monthly Niño 3.4 sea surface temperature anomaly, $\overline{SST'}$ is the mean of the anomaly series, and $\sigma_{SST'}$ is its standard deviation. The standardized Niño 3.4 index is treated as a continuous measure of ENSO conditions throughout the statistical analysis.

3.5 Statistical Analysis

The historical relationship between ENSO and rainfall is evaluated using the Pearson correlation coefficient. Correlations are calculated between the standardized Niño 3.4 index and both precipitation anomalies and SPI.

For each relationship, three statistics are reported: the Pearson correlation coefficient r , the corresponding p , and r^2 . The correlation coefficient measures the direction and strength of the linear relationship, while the p -value is used to evaluate statistical significance. The squared correlation coefficient r^2 provides a measure of the magnitude of the linear association.

Correlations are estimated at 0, 1, 3 and 6 month lags to examine whether rainfall is associated with ENSO conditions in the same month or in preceding months. The regional analysis is complemented by a grid-cell analysis of the relationship between ENSO and SPI-6 across Vietnam.

4. How Unusual Was Rainfall in Vietnam in 2025?

4.1 Historical Rainfall Context

Rainfall in Vietnam is highly uneven across space, reflecting differences in latitude, topography, monsoon exposure and proximity to the coast. Before assessing whether 2025 was unusual, it is therefore necessary to establish the historical rainfall pattern against which recent conditions are compared.

Figure 4 shows the climatological distribution of annual precipitation across Southeast Asia. The region is generally characterized by high rainfall, with the highest annual totals concentrated across maritime Southeast Asia, particularly Borneo, Sumatra, Sulawesi and Papua. Vietnam also shows substantial spatial variation in annual rainfall, which is examined in greater detail in the following figure.

A more detailed view of Vietnam is shown in **Figure 5**. The map highlights substantial regional differences in climatological annual precipitation across the country. Particularly high rainfall is evident around central Vietnam, including the area around Hue, where the Annamite Range and Bach Ma Mountains enhance rainfall through orographic uplift. South of this area, the rainfall pattern changes markedly, reflecting differences in topography and seasonal monsoon exposure. These regional contrasts provide an important baseline for assessing how unusual rainfall conditions were in 2025.

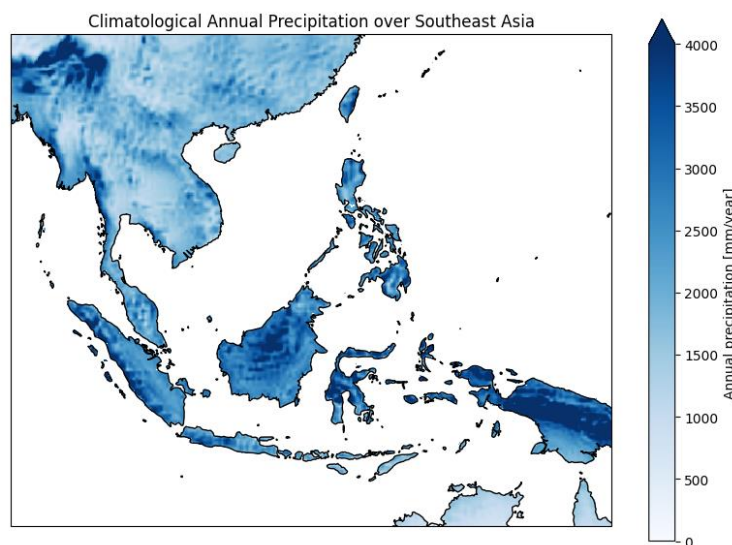


Figure 4 Climatological Annual Precipitation over Southeast Asia

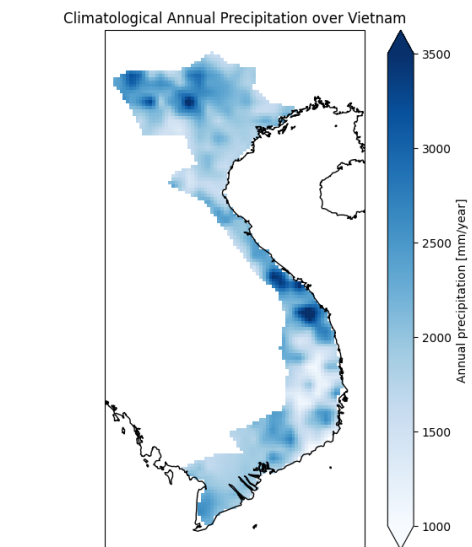


Figure 5. Climatological Annual Precipitation over Vietnam

4.2 Annual Rainfall Anomalies in Two Wet Years: 2024 and 2025

Figure 6 compares annual precipitation anomalies across the six regions in 2024 and 2025. Both years were wetter than the historical average in most regions, making 2024 a useful recent benchmark for assessing how exceptional 2025 was. The comparison shows that rainfall anomalies increased sharply in several central regions in 2025, especially North Central Vietnam, South Central Vietnam and the Central Highlands. **Figure 7** complements this annual comparison by showing the monthly evolution of precipitation across the same six regions in both years, making it easier to see how rainfall developed through 2024 and 2025.

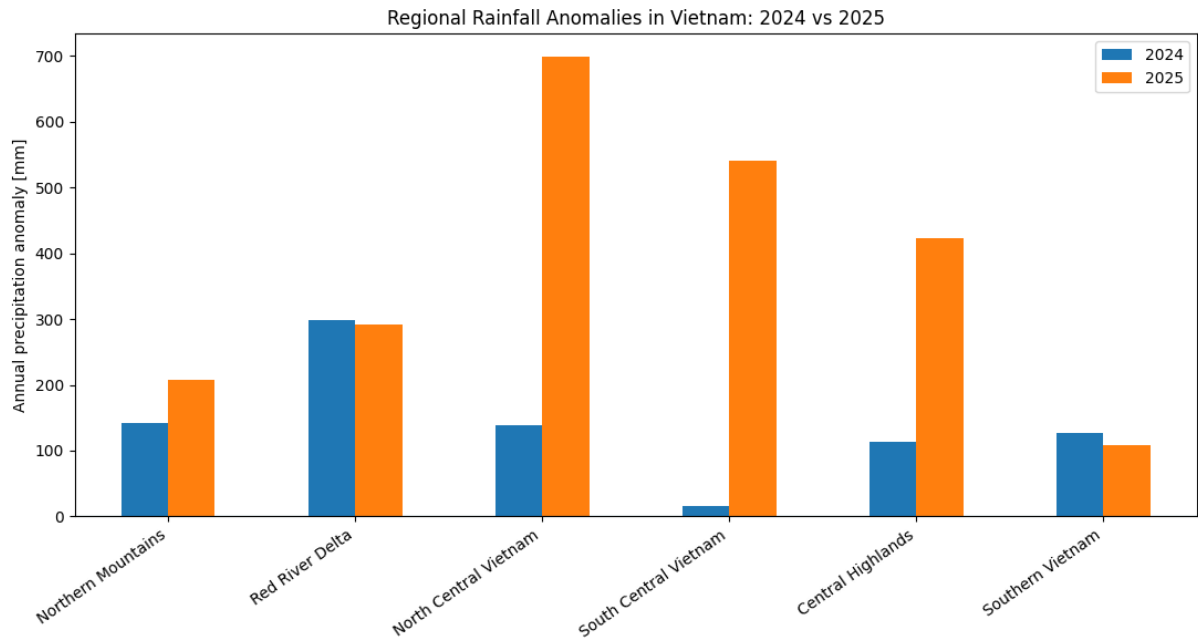


Figure 6. Regional Annual Rainfall Anomalies in Vietnam: 2024 vs 2025



Figure 7. Monthly Precipitation by Region in Vietnam: 2024 vs 2025

Table 1. Regional Annual Precipitation Anomalies in 2024 and 2025

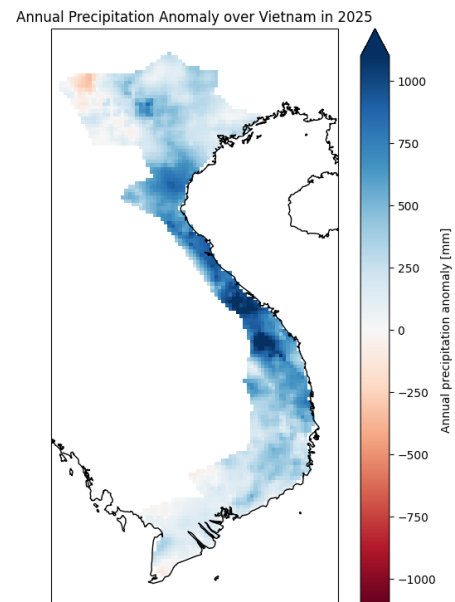
Region	2024 anomaly (mm)	2025 anomaly (mm)
Northern Mountains	+141.8	+206.8
Red River Delta	+298.7	+291.2
North Central Vietnam	+139.4	+698.9
South Central Vietnam	+15.1	+540.6
Central Highlands	+113.1	+422.3
Southern Vietnam	+127.2	+108.0

4.3 Spatial Distribution of the 2025 Rainfall Anomaly

Regional averages can conceal substantial variation within each region. **Figure 8** therefore shows the spatial distribution of the annual precipitation anomaly across Vietnam in 2025.

The map reveals a clear concentration of positive rainfall anomalies across central Vietnam, forming a broad wet corridor from North Central into South Central Vietnam and extending into parts of the Central Highlands. The strongest anomalies are clustered within this central portion of the country, while northern and southern Vietnam show generally weaker and more mixed departures from their historical rainfall levels. The pattern therefore indicates that the unusually wet conditions of 2025 were strongly concentrated in central Vietnam rather than being distributed uniformly across the country.

Figure 8. Annual Precipitation Anomaly over Vietnam in 2025



5. Historical Relationship Between ENSO and Rainfall

The analysis now turns from describing rainfall conditions in 2025 to examining the historical relationship between ENSO and precipitation across Vietnam. The objective is to assess whether a consistent ENSO signal is visible at the national level, how this relationship varies across regions and accumulation periods, and whether it changes when lagged ENSO conditions are considered.

5.1 National ENSO–SPI Relationship

Figure 9 compares the historical ENSO Index with the national SPI-6 series for Vietnam. The national SPI-6 series is constructed from the area-weighted spatial mean of grid-cell SPI-6 values and standardized again after spatial aggregation. The two series show a broadly negative relationship over time: when the ENSO Index rises to relatively high values, SPI-6 often moves toward lower values, while periods of relatively low ENSO values frequently coincide with higher SPI-6 values. In other words, the two series tend to move in opposite directions.

The climatic interpretation follows from what the two indices represent. Higher ENSO values indicate relatively warmer sea surface temperatures in the Niño 3.4 region, while higher SPI-6 values indicate wetter precipitation conditions in Vietnam over the preceding six months. The negative relationship therefore means that relatively warmer ENSO conditions tend to be associated with drier conditions in Vietnam, while relatively cooler ENSO conditions tend to be associated with wetter conditions. This

pattern is broadly consistent with the ENSO mechanism described in Section 2. During El Niño conditions, warmer water and convection shift toward the central and eastern Pacific, which can reduce rainfall across parts of Southeast Asia and the western Pacific. During La Niña conditions, the pattern broadly reverses and wetter conditions become more likely in this region. The inverse relationship appears repeatedly over the historical period, but not at every point in time, indicating that ENSO captures only part of Vietnam's precipitation variability.

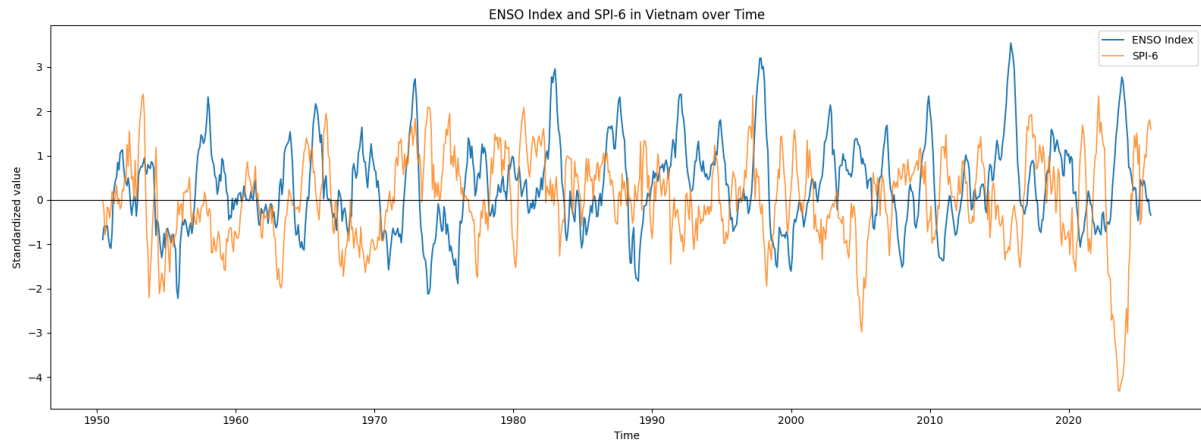


Figure 9. ENSO Index and SPI-6 in Vietnam over Time

5.2 Regional ENSO and Rainfall Relationship

Table 2 reports the contemporaneous correlation between the ENSO Index and monthly precipitation anomalies across the six regions. The relationship is negligible in the Northern Mountains and Red River Delta, but becomes more clearly negative across central and southern Vietnam. The strongest contemporaneous correlation is found in South Central Vietnam, followed by Southern Vietnam, the Central Highlands and North Central Vietnam.

Table 2. Regional Correlation between ENSO and Precipitation Anomalies

Region	Pearson r	r^2 (%)
Northern Mountains	0.002	0.0
Red River Delta	-0.022	0.0
North Central Vietnam	-0.148***	2.2
South Central Vietnam	-0.262***	6.9
Central Highlands	-0.179***	3.2
Southern Vietnam	-0.213***	4.6

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The regional results show that the ENSO–rainfall relationship is much more apparent across central and southern Vietnam than in the north. South Central Vietnam shows the strongest contemporaneous relationship, with a correlation of approximately -0.26, while the Northern Mountains and Red River Delta show little relationship with ENSO.

5.3 ENSO Relationship across SPI Time Scales

The relationship is next examined using SPI at different accumulation periods because ENSO may be associated more strongly with rainfall conditions that persist over several months than with rainfall in a single month. Different SPI time scales capture different types of precipitation variability. SPI-1 reflects short-term monthly conditions and is sensitive to individual wet or dry months. SPI-3 represents precipitation accumulated over roughly one season, while SPI-6 captures more persistent wet or dry conditions developing over several months. SPI-24 reflects much longer-term precipitation conditions and changes more gradually.

Comparing these accumulation periods therefore helps identify the time scale at which the ENSO signal is most clearly reflected in rainfall. **Table 3** shows that the negative relationship generally becomes stronger at SPI-3 and SPI-6 across central and southern Vietnam. The strongest relationship is found for SPI-6 in South Central Vietnam, where the correlation reaches approximately -0.39, corresponding to about 15% of the linear variation.

Table 3. Correlation between ENSO and SPI across Accumulation Periods

Region	SPI-1	SPI-3	SPI-6	SPI-24
Northern Mountains	0.037	0.018	0.005	0.081*
Red River Delta	0.054	0.022	-0.041	0.041
North Central Vietnam	-0.126***	-0.234***	-0.253***	-0.082*
South Central Vietnam	-0.223***	-0.342***	-0.392***	-0.180***
Central Highlands	-0.187***	-0.252***	-0.230***	-0.077*
Southern Vietnam	-0.242***	-0.296***	-0.297***	-0.091**

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The stronger correlations at SPI-3 and SPI-6 indicate that the ENSO relationship is more visible when precipitation is considered over seasonal to multi-month periods. SPI-1 contains more short-term rainfall variability, while SPI-24 combines precipitation over a much longer period in which the ENSO signal becomes less distinct. The results therefore suggest that the historical ENSO–rainfall relationship in Vietnam is most apparent at intermediate accumulation periods of approximately three to six months.

5.4 Timing of the ENSO Relationship

The ENSO–rainfall relationship may not occur only within the same month. Changes in tropical Pacific conditions can be followed by changes in atmospheric circulation and rainfall over subsequent months, so it is useful to examine whether earlier ENSO conditions are associated with later precipitation conditions. Lagged analysis therefore helps identify whether the ENSO signal is strongest contemporaneously or remains visible after one, three or six months. **Table 4** compares monthly precipitation anomalies with ENSO values observed at these different lags.

Table 4. Lagged Regional Correlation between ENSO and Precipitation Anomalies

Region	Lag 1 month	Lag 3 months	Lag 6 months
Northern Mountains	0.002	-0.011	-0.025
Red River Delta	-0.016	-0.030	-0.028

Region	Lag 1 month	Lag 3 months	Lag 6 months
North Central Vietnam	-0.119***	-0.110***	-0.064
South Central Vietnam	-0.222***	-0.182***	-0.065
Central Highlands	-0.149***	-0.155***	-0.064
Southern Vietnam	-0.188***	-0.189***	-0.064

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The negative relationship remains visible across central and southern Vietnam at one- and three-month lags, while it weakens substantially by six months. This suggests that the ENSO signal is most apparent within the same month and the following few months. The Northern Mountains and Red River Delta continue to show little relationship across the tested lags.

5.5 Spatial Pattern of the ENSO–SPI Relationship

Regional averages can conceal substantial variation within each region. **Figure 10** therefore maps the correlation between the ENSO Index and SPI-6 at each grid cell across Vietnam, showing where the historical relationship is strongest and where it is weak.

The spatial pattern is predominantly negative across central and southern Vietnam, with many statistically significant grid cells, while northern Vietnam shows weaker and more mixed relationships, including some areas with positive correlations. Approximately 71% of valid grid cells are statistically significant at the 5% level. The stronger negative pattern across central and southern Vietnam is broadly consistent with the regional results reported above.

These spatial differences may reflect the way ENSO interacts with other components of Vietnam's climate system. Rainfall in central and southern Vietnam is strongly influenced by tropical Pacific circulation, monsoon variability and moisture transport, allowing the ENSO signal to appear more clearly. In northern Vietnam, rainfall is also affected by the East Asian monsoon, cold surges, topography and other regional weather systems, which can weaken or modify the ENSO relationship. The map therefore shows that the influence of ENSO on rainfall varies substantially across Vietnam.

Spatial Correlation between ENSO and SPI-6 across Vietnam

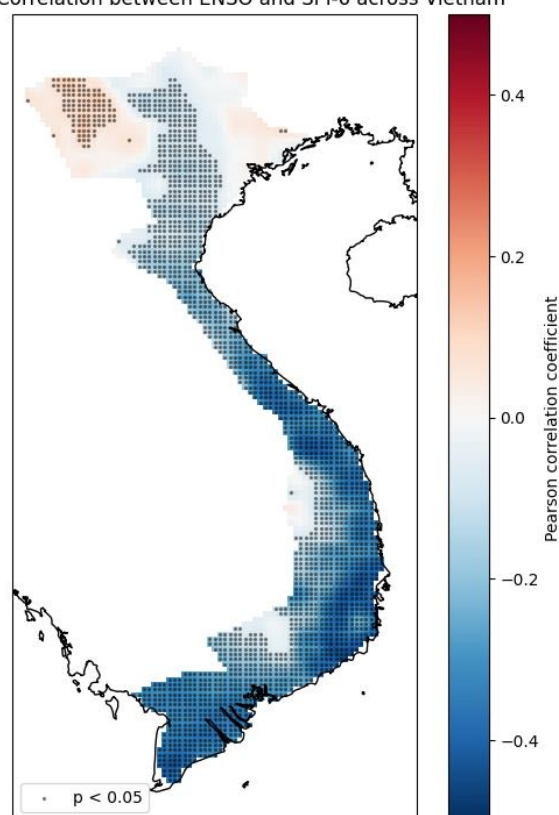


Figure 10. Spatial Correlation between ENSO and SPI-6 across Vietnam

Overall, the historical analysis shows that ENSO has a clear but uneven relationship with rainfall in Vietnam. The negative relationship is most visible across central and southern Vietnam, becomes stronger when precipitation is accumulated over three to six months, and remains detectable at short lags. The spatial analysis further shows that this relationship varies substantially across the country.

6. Was 2025 Consistent With the Historical ENSO Signal?

Section 5 showed that ENSO has a statistically significant relationship with precipitation across much of central and southern Vietnam, with the relationship becoming clearer when rainfall is accumulated over several months. In particular, the historical correlations were stronger for SPI-3 and SPI-6 than for monthly precipitation in several regions. This section uses those historical relationships as a benchmark to assess whether precipitation conditions in 2025 were consistent with the rainfall conditions normally associated with ENSO.

6.1 ENSO Conditions in 2025

ENSO conditions changed during 2025. **Table 5** reports the standardized monthly Niño 3.4 SST anomaly used in this study together with the corresponding ENSO phase. The standardized index measures the relative monthly Niño 3.4 SST anomaly used in the statistical analysis, while the ENSO phase reflects the broader coupled ocean and atmosphere state.

Table 5. Standardized Niño 3.4 SST Anomaly and ENSO Phase in 2025

Month	ENSO Index	ENSO Phase
January	-0.47	La Niña
February	-0.07	La Niña
March	0.45	Neutral
April	0.36	Neutral
May	0.31	Neutral
June	0.44	Neutral
July	0.35	Neutral
August	0.09	Neutral
September	-0.04	La Niña
October	0.01	La Niña
November	-0.26	La Niña
December	-0.34	La Niña

Note: The standardized ENSO Index is used for the statistical analysis and is not itself used to classify ENSO phases. ENSO phase classifications follow NOAA operational monitoring, which considers the broader coupled ocean–atmosphere state.

La Niña conditions were present in January and February, followed by ENSO-neutral conditions from March through August. La Niña conditions returned from September through December. The ENSO Index also changed substantially within these phases, ranging from -0.47 in January to +0.45 in March before moving back into negative territory toward the end of the year. The evolution of ENSO during 2025 therefore needs to be considered month by month when comparing it with rainfall conditions.

6.2 Comparing 2025 with the Historical ENSO–SPI Relationship

The 2025 comparison focuses on SPI-3 and SPI-6, which showed clearer historical relationships with ENSO than monthly precipitation in several regions. For each region, the historical ENSO–SPI

relationship over 1950–2024 is used to estimate the SPI conditions normally associated with the observed ENSO Index. The difference between observed and expected SPI is then used to measure whether 2025 was wetter or drier than the historical relationship would suggest.

Because consecutive SPI-3 and SPI-6 observations contain overlapping months of rainfall, they are not statistically independent. A moving block bootstrap is therefore used to preserve this temporal dependence. Three-month blocks are used for SPI-3 and six-month blocks for SPI-6. The mean difference between observed and expected SPI across the twelve months of 2025 is then compared with the corresponding historical bootstrap distribution.

6.3 Regional Results for 2025

Table 6 reports the mean ENSO-adjusted SPI residual in 2025 together with the historical 95% bootstrap range and two-sided bootstrap p-value. Positive residuals indicate wetter conditions than expected from the historical ENSO relationship.

Table 6. ENSO-Adjusted SPI Conditions in Vietnam in 2025

Region	SPI-3 Mean Residual	SPI-3 95% Bootstrap Range	SPI-6 Mean Residual	SPI-6 95% Bootstrap Range
Northern Mountains	0.05	-0.83 to 0.83	0.11	-1.23 to 1.06
Red River Delta	0.13	-0.83 to 0.83	0.37	-1.20 to 1.18
North Central Vietnam	1.10**	-0.79 to 0.80	1.39*	-1.16 to 1.20
South Central Vietnam	0.94*	-0.78 to 0.81	0.97	-1.04 to 1.10
Central Highlands	0.82	-0.82 to 0.84	0.99	-1.19 to 1.19
Southern Vietnam	0.34	-0.85 to 0.83	0.49	-1.21 to 1.09

Note: Mean residuals are calculated as observed SPI minus the SPI expected from the historical ENSO–SPI relationship over 1950–2024. The 95% ranges are estimated using a moving block bootstrap with three-month blocks for SPI-3 and six-month blocks for SPI-6. Significance is based on two-sided bootstrap p-values. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The clearest departure is found in North Central Vietnam. The mean SPI-3 residual is 1.10, exceeding the upper 95% bootstrap bound of 0.80. The SPI-6 residual is also outside the historical range at 1.39, compared with an upper bound of 1.20. The unusually wet conditions in North Central Vietnam therefore remain evident at both seasonal and six-month accumulation periods.

South Central Vietnam also shows wetter conditions than expected from the historical ENSO relationship. The SPI-3 mean residual is 0.94, above the upper bootstrap bound of 0.81, while the SPI-6 residual of 0.97 remains within the corresponding historical range. The departure is therefore clearer at the seasonal scale than at the longer six-month accumulation period.

The Northern Mountains, Red River Delta and Southern Vietnam show no statistically significant departures at either accumulation period. Taken together, the regional results indicate that the strongest deviations from the historical ENSO signal in 2025 were concentrated in central Vietnam, particularly North Central Vietnam.

6.4 Interpretation

The 2025 results show that ENSO provides useful information about broad rainfall variability, but it does not fully account for the wet conditions observed across central Vietnam. North Central Vietnam shows the clearest departure, with precipitation conditions significantly wetter than the historical ENSO benchmark at both SPI-3 and SPI-6. South Central Vietnam also shows an unusual seasonal wet signal at SPI-3, although the departure is weaker at the six-month scale.

Overall, 2025 was only partly consistent with the historical ENSO signal. The historical ENSO relationship remains visible, but the magnitude of wet conditions in parts of central Vietnam exceeded what would normally be expected from ENSO alone. The following case studies examine Hue and Khanh Hoa in greater detail to assess how these regional patterns appeared at specific locations and during individual rainfall episodes.

7. Case Studies

The regional analysis identifies central Vietnam as the main area in which rainfall conditions during 2025 departed from the historical ENSO signal. The following case studies examine these conditions at a more local scale. Each case first evaluates how rainfall differed from its historical climatology, then places the most unusual month within its historical distribution, and finally compares the event with the historical ENSO–rainfall relationship.

7.1 Hue, October 2025

Hue experienced several unusually wet months during 2025. **Figure 11** shows monthly precipitation anomalies relative to the corresponding historical monthly climatology. Positive anomalies occurred in several months, with particularly large departures in June, August and October. October stands out most clearly, with a precipitation anomaly of approximately +464 mm above its historical October average.

October rainfall reached approximately 931.9 mm, compared with a historical October mean of 467.7 mm. To assess how unusual this was, **Figure 12** places the October 2025 observation within the historical distribution of October rainfall. The observation lies well into the upper tail of the distribution, corresponding to approximately the 97.3rd percentile and ranking as the third-wettest October among 76 observations from 1950 to 2025. Based on its empirical rank, this corresponds to a return period of 25.7 years.

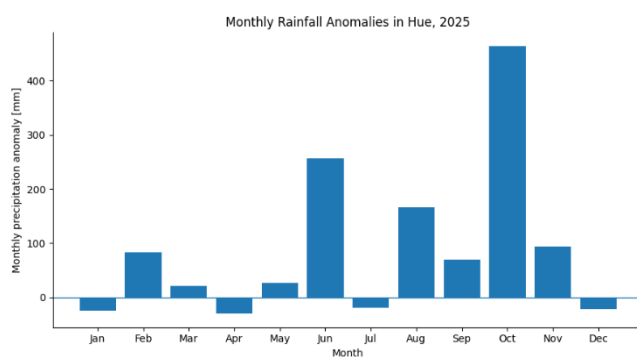


Figure 11. Monthly Rainfall Anomalies in Hue, 2025

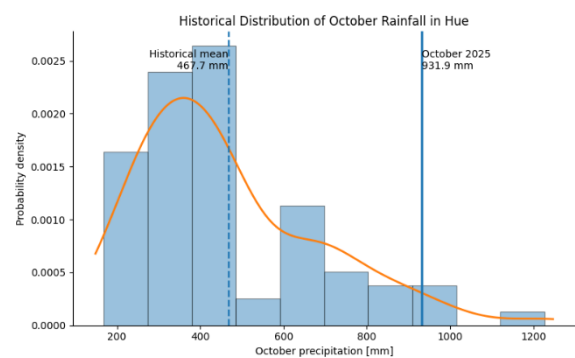


Figure 12. Historical Distribution of October Rainfall in Hue

The historical ENSO relationship provides additional context for the event. **Table 7** compares October rainfall in Hue with contemporaneous and earlier Niño 3.4 conditions. The historical correlations are negative at the contemporaneous, one-month and three-month leads, indicating that warmer Niño 3.4 conditions have generally been associated with lower October rainfall in Hue. However, the relationship is relatively limited even contemporaneously, where $r=-0.218$ and $r^2=4.7\%$. ENSO therefore captures only a small part of the historical variation in October rainfall at this location.

Table 7. Historical ENSO Relationship and ENSO Conditions for October 2025 Rainfall in Hue

ENSO lead	Reference month	Pearson r	r^2 (%)	Niño 3.4	Interpretation
0 months	October 2025	-0.218***	4.7	+0.01	Neutral ENSO
1 month	September 2025	-0.179***	3.2	-0.04	Neutral ENSO
3 months	July 2025	-0.142***	2.0	+0.35	Direction inconsistent with wetter conditions
6 months	April 2025	-0.049	0.2	+0.36	Historical relationship not significant

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The contemporaneous Niño 3.4 index in October was +0.01, while the September value was -0.04, placing both months close to the historical mean. ENSO therefore provided little signal of the exceptionally wet conditions observed in October. At a three-month lead, the July Niño 3.4 index was positive at +0.35. Given the historically negative ENSO–rainfall relationship, this direction would normally be associated with lower rainfall and is therefore inconsistent with the observed October rainfall. At a six-month lead, the historical relationship is not statistically significant.

Taken together, the results show that ENSO alone provides only limited information about an extreme monthly rainfall event such as October 2025 in Hue. Even the strongest historical relationship accounts for only about 4.7% of the linear variation in October rainfall, while ENSO conditions immediately before and during the event were close to neutral. The exceptionally high rainfall therefore reflects substantial variability beyond the large-scale ENSO signal, pointing to the importance of other regional and local weather processes in producing individual rainfall extremes. This is consistent with the broader finding for North Central Vietnam in Section 6, where 2025 was substantially wetter than the level implied by the historical ENSO relationship.

7.2 Khanh Hoa, November 2025

Khanh Hoa also experienced several wetter-than-normal months during 2025, although the pattern was less extreme than in Hue. **Figure 13** shows monthly precipitation anomalies relative to the historical monthly climatology. Positive anomalies were particularly visible in February, August, September and November, with November showing the largest departure at approximately +192 mm.

November rainfall reached approximately 475.2 mm, compared with a historical November mean of 283.5 mm. **Figure 14** places the 2025 observation within the historical distribution of November rainfall. The observation corresponds to approximately the 89.3rd percentile and ranks as the ninth-wettest November among 76 observations from 1950 to 2025. Based on its empirical rank, this corresponds to a return period of 8.6 years.

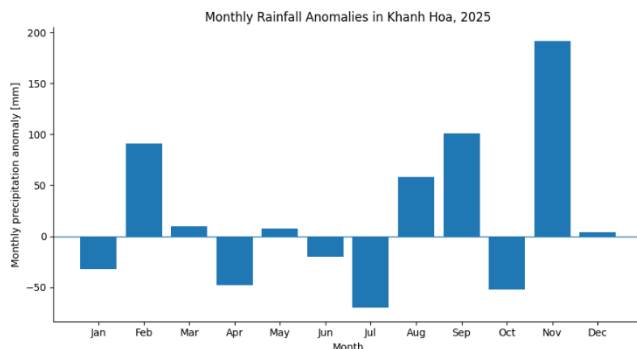


Figure 13. Monthly Rainfall Anomalies in Khanh Hoa, 2025

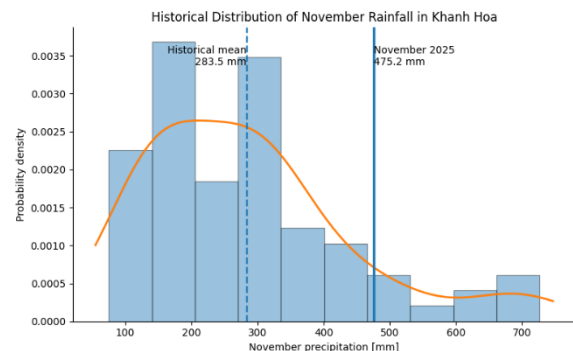


Figure 14. Historical Distribution of November Rainfall in Khanh Hoa

The historical ENSO relationship is again negative at the shorter leads. As shown in **Table 8**, the contemporaneous relationship is the strongest, with $r=-0.236$, corresponding to $r^2=5.6\%$. The relationship remains statistically significant at one- and three-month leads but weakens with increasing lead time.

Table 8. Historical ENSO Relationship and ENSO Conditions for November 2025 Rainfall in Khanh Hoa

ENSO lead	Reference month	Pearson r	r ² (%)	Niño 3.4	Interpretation
0 months	November 2025	-0.236***	5.6	-0.26	Direction consistent with wetter conditions
1 month	October 2025	-0.198***	3.9	+0.01	Neutral ENSO
3 months	August 2025	-0.168***	2.8	+0.09	Neutral ENSO
6 months	May 2025	-0.050	0.2	+0.31	Historical relationship not significant

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The contemporaneous Niño 3.4 index in November was -0.26. Given the historically negative ENSO–rainfall relationship, this direction is consistent with wetter conditions in Khanh Hoa. However, the magnitude of the historical relationship remains limited, with contemporaneous ENSO accounting for only about 5.6% of the linear variation in November rainfall. The October and August ENSO values were close to the historical mean, while the six-month historical relationship is not statistically significant.

Taken together, November 2025 was clearly wetter than normal in Khanh Hoa, but it was less historically unusual than the October event in Hue. The contemporaneous ENSO signal was directionally consistent with wetter conditions, although its historical relationship with rainfall is relatively weak. ENSO may therefore have provided part of the large-scale background associated with the event, but it cannot account for the magnitude of November rainfall on its own. This is also consistent with the regional results for South Central Vietnam in Section 6, where 2025 showed an unusual wet signal at the seasonal SPI-3 scale but weaker evidence at SPI-6.

7.3 Ho Chi Minh City, July 2026

Ho Chi Minh City is included as an additional case outside the main 2025 analysis. The available data currently cover January to August 2026, allowing the first eight months of the year to be compared with their historical monthly climatology. **Figure 15** shows that rainfall anomalies were relatively modest across most months. July stands out as the wettest departure within the available period, with rainfall approximately 52 mm above its historical July average.

July rainfall reached approximately 328.3 mm, compared with a historical July mean of 276.2 mm. **Figure 16** places this observation within the historical distribution of July rainfall. The July 2026 observation corresponds to approximately the 84.2nd percentile and ranks as the 13th-wettest July among 77 observations. Based on its empirical rank, this corresponds to an approximate return period of 6.0 years. The event was therefore wetter than normal, but not particularly rare within the historical July distribution.

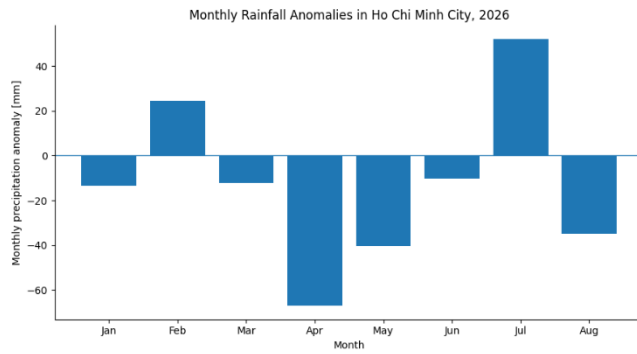


Figure 15. Monthly Rainfall Anomalies in Ho Chi Minh City, 2026

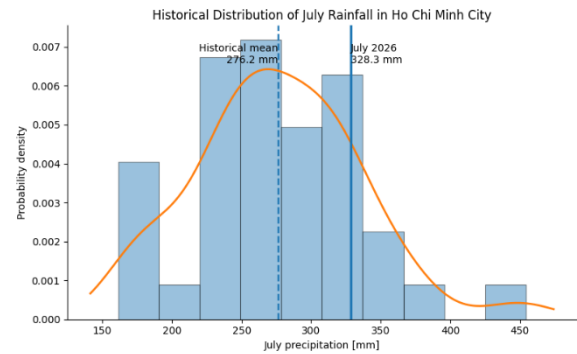


Figure 16. Historical Distribution of July Rainfall in Ho Chi Minh City

The historical ENSO relationship is negative at all four tested leads, although its magnitude is limited. **Table 9** shows that the contemporaneous relationship has $r=-0.192$, corresponding to $r^2=3.7\%$. The one-month and three-month relationships are of similar magnitude, while the six-month relationship is considerably weaker.

Table 9. Historical ENSO Relationship and ENSO Conditions for July 2026 Rainfall in Ho Chi Minh City

ENSO lead	Reference month	Pearson r	r^2 (%)	Niño 3.4	Interpretation
0 months	July 2026	-0.192***	3.7	+2.73	Direction inconsistent with wetter conditions
1 month	June 2026	-0.164***	2.7	+2.23	Direction inconsistent with wetter conditions
3 months	April 2026	-0.174***	3.0	+0.85	Direction inconsistent with wetter conditions
6 months	January 2026	-0.069*	0.5	-0.29	Direction consistent with wetter conditions

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The Niño 3.4 index was strongly positive in July at +2.73 and had already been positive in June and April. Given the historically negative ENSO–rainfall relationship, these conditions would normally be associated with lower rainfall in Ho Chi Minh City. The observed positive July rainfall anomaly therefore runs in the opposite direction from the historical ENSO signal at contemporaneous, one-month and three-month leads. At the six-month lead, the direction is consistent with wetter conditions, but the historical relationship is much weaker.

Taken together, July 2026 was wetter than normal in Ho Chi Minh City, but it was not a historically rare rainfall month. The event also occurred during strongly positive Niño 3.4 conditions, which were directionally inconsistent with the historical ENSO–rainfall relationship at the shorter leads. As in the Hue and Khanh Hoa cases, ENSO alone provides limited information about the magnitude of an individual monthly rainfall event.

The available dataset currently ends in August 2026. Rainfall in Ho Chi Minh City appears to have been substantial again in September based on more recent observations and reports, but September is not included in the present analysis because the corresponding ERA5-Land monthly data were not yet available in the dataset used for this study.

8. Discussion

The analysis reveals several consistent patterns in the relationship between ENSO and rainfall in Vietnam. First, rainfall conditions in 2025 were not uniformly unusual across the country. The strongest positive anomalies were concentrated in central Vietnam, particularly in North Central and South Central Vietnam. This concentration matters because the unusually wet conditions of 2025 were mainly a regional phenomenon, not a uniform pattern across Vietnam.

The historical ENSO analysis also shows that the relationship depends strongly on the time scale over which precipitation is measured. ENSO correlations are relatively weak for SPI-1, become stronger at SPI-3 and SPI-6, and weaken again at SPI-24. This pattern suggests that ENSO is most closely associated with seasonal to multi-month rainfall variability. Monthly rainfall contains substantial short-term variability, while very long accumulation periods combine precipitation over time scales that are less closely aligned with the ENSO signal.

This temporal pattern is consistent with the lagged analysis. Across central and southern Vietnam, the negative ENSO-rainfall relationship remains visible at contemporaneous, one-month and three-month lags, but weakens substantially by six months. The historical relationship therefore appears to operate most clearly over the same season and the following few months. ENSO provides useful information about the broader rainfall background over these time scales, but its value declines when it is treated as a longer-horizon predictor.

The relationship also varies considerably across space. The Northern Mountains and Red River Delta show little historical association with ENSO, while the relationship becomes more clearly negative across central and southern Vietnam. The grid-cell analysis confirms this pattern, showing predominantly negative correlations across much of central and southern Vietnam and weaker, more mixed relationships in the north. This spatial heterogeneity indicates that the ENSO-rainfall relationship should be interpreted regionally rather than through a single national response.

These historical patterns provide the benchmark for interpreting 2025. The year broadly fits the established spatial structure in which ENSO is more relevant for central and southern Vietnam than for the north, but the magnitude of rainfall in parts of central Vietnam exceeded what the historical ENSO relationship would normally imply. North Central Vietnam stands out most clearly, with significantly positive ENSO-adjusted residuals at both SPI-3 and SPI-6. South Central Vietnam also shows an unusual positive departure at SPI-3, although the SPI-6 result is less statistically distinct. The strongest departures in 2025 were therefore concentrated in central Vietnam, where rainfall was wetter than the historical ENSO relationship alone would suggest.

This distinction is important. The historical results show that ENSO can help describe the background tendency toward wetter or drier seasonal conditions, particularly over three- to six-month periods. The 2025 results show the limits of that relationship. Even where the ENSO signal is statistically clear, a substantial part of rainfall variability remains outside the historical ENSO relationship. ENSO should therefore be interpreted as one large-scale climate influence within a broader set of regional and local rainfall processes.

The case studies extend this interpretation beyond rainfall variability alone. Hue and Khanh Hoa can be considered primarily in the context of fluvial flood risk, where heavy rainfall may translate into flooding through river and catchment systems, while Ho Chi Minh City provides a more pluvial flood setting, where intense rainfall can generate surface flooding when urban drainage capacity is exceeded. This distinction matters because rainfall magnitude alone does not determine the severity of flood impacts.

The contrast between Hue and Khanh Hoa is particularly informative. October 2025 rainfall in Hue was highly unusual, ranking as the third-wettest October in the historical record. November rainfall in Khanh Hoa was clearly above normal but much less rare, with an approximate return period of 8.6 years.

Despite this, the event was still associated with substantial flooding and disruption. This suggests that the rarity of rainfall alone is not sufficient to explain the severity of a flood event.

A broader interpretation therefore requires considering hazard, exposure and vulnerability together. Rainfall represents an important component of the hazard, but the eventual impact also depends on how rainfall translates into flooding, which people and assets are exposed, and how vulnerable they are when flooding occurs. River-basin geography, drainage systems, settlement patterns, infrastructure and preparedness may therefore help explain why rainfall events of different statistical rarity can produce severe impacts. These factors are not quantified in the present study and provide an important direction for further research.

Overall, the results suggest a clear hierarchy in how ENSO should be interpreted in Vietnam. ENSO is most informative for regional, seasonal and multi-month rainfall variability, while its relationship with individual monthly extremes is considerably weaker. The 2025 experience also shows that understanding rainfall is only one part of understanding flood risk. Once the analysis moves from rainfall to actual damages, local flood mechanisms, exposure and vulnerability become increasingly important.

9. Limitations and Future Research

9.1 Limitations

Several limitations should be considered when interpreting the results. First, ENSO is only one component of the climate system influencing rainfall in Vietnam. Monsoon circulation, tropical disturbances, moisture transport, topography and other regional atmospheric processes can all affect precipitation, particularly at shorter time scales. This is especially important for the case studies, where the historical ENSO relationship explains only a limited share of monthly rainfall variation.

Second, the statistical analysis mainly captures linear relationships. Pearson correlation and the ENSO-based regression framework are useful for identifying broad historical associations, but they may not fully capture nonlinear responses, threshold effects or interactions between ENSO and other climate drivers. The lag analysis also considers only selected lead periods, so the results should be interpreted as evidence of broad temporal patterns rather than a complete representation of rainfall dynamics.

The SPI analysis introduces another limitation. SPI-3 and SPI-6 are useful for identifying seasonal and multi-month rainfall conditions, but their overlapping accumulation windows create serial dependence between consecutive observations. The moving block bootstrap used in Section 6 preserves this temporal dependence when evaluating the 2025 episode, but the framework remains a simplified representation of the underlying time-series dynamics.

The precipitation data also impose an important limitation because the present study uses monthly rainfall totals. This temporal resolution is appropriate for analysing seasonal rainfall variability, SPI and historical ENSO relationships, but it is too coarse for detailed event analysis or flood-risk modelling. Monthly totals cannot distinguish between rainfall accumulated gradually over several weeks and an intense rainfall episode concentrated within a few hours or days. Higher-frequency data, such as daily or hourly precipitation, would therefore be needed to analyse individual flood events and short-duration rainfall extremes more accurately.

The case-study results should therefore be interpreted primarily as analyses of monthly rainfall conditions, not as formal attribution of individual flood events. A complete flood-risk assessment would require more detailed information on rainfall intensity, river response, drainage, flood depth and extent, and the spatial distribution of exposed assets and populations.

9.2 Future Research

A first extension would be to improve the hazard analysis using higher-frequency precipitation data together with a broader set of climate and atmospheric variables. Daily rainfall data would allow individual wet spells and event sequences to be identified, while hourly or sub-daily data would be more suitable for analysing short-duration rainfall intensity and pluvial flooding. These data could be combined with monsoon indices, tropical cyclone activity, wind fields, atmospheric moisture transport and pressure patterns to examine how ENSO interacts with other large-scale and regional processes in producing rainfall extremes. More flexible statistical or time-series models could then be used to capture nonlinear relationships and interactions between these drivers.

A second major extension would be to move from rainfall analysis toward a broader flood-risk framework based on hazard, exposure and vulnerability. The hazard component would describe the probability and intensity of flooding, including differences between fluvial and pluvial processes. Exposure would identify the households, firms, infrastructure and other assets located within affected areas, while vulnerability would describe how strongly those exposed assets are damaged when flooding occurs. For fluvial flood risk, this could include river networks, catchment characteristics, upstream rainfall, flood depth and extent, and the location of settlements and infrastructure within flood-prone areas. For pluvial flood risk, particularly in urban areas such as Ho Chi Minh City, relevant variables could include short-duration rainfall intensity, drainage capacity, impervious surfaces, land use and local runoff conditions. Combining these components would allow future research to estimate physical flood losses and then connect those losses to financial and economic models. For banks, this could involve borrower default risk, collateral values and credit losses. For insurers, the analysis could examine claims, insured losses and changes in risk pricing. At a broader level, physical losses could be incorporated into regional or macroeconomic models to study effects on household consumption, business activity, investment, public expenditure and economic output.

A longer-term research framework could therefore follow the sequence:

climate signal → rainfall and flood hazard → exposure → vulnerability → physical losses → financial and macroeconomic impacts

The present study focuses on the first part of this chain by examining rainfall variability, ENSO relationships and the unusually wet conditions observed in central Vietnam. Extending the analysis along this framework would allow the climate findings to be translated into measures that are more directly relevant for financial institutions, insurers, policymakers and climate adaptation planning.

10. Conclusion

This study examined how unusual rainfall conditions were in Vietnam in 2025 and the extent to which ENSO could help explain the observed variability. The results show that the wet conditions of 2025 were not uniform across the country. The strongest positive rainfall anomalies were concentrated in central Vietnam, particularly in North Central and South Central Vietnam.

The historical analysis also shows that the ENSO–rainfall relationship varies across both space and time scale. The relationship is weak in northern Vietnam, stronger across central and southern Vietnam, and most apparent when precipitation is accumulated over three to six months. ENSO is therefore more informative for seasonal and multi-month rainfall variability than for individual monthly extremes.

When the 2025 observations are compared with the historical ENSO relationship, the year appears only partly consistent with the historical signal. North Central Vietnam was significantly wetter than the ENSO-based historical benchmark at both SPI-3 and SPI-6, while South Central Vietnam showed a significant positive departure at SPI-3 but not at SPI-6. ENSO therefore provides useful information

about the broader rainfall background, but it does not fully account for the magnitude of the wet conditions observed in central Vietnam.

The case studies reinforce this conclusion at the local scale. Rainfall rarity and ENSO conditions alone are not sufficient to explain the severity of flood impacts. Understanding flood risk requires considering how rainfall translates into flooding and how that hazard interacts with exposure and vulnerability. The present study therefore provides a first step toward a broader climate-risk framework in which climate signals and rainfall hazards can later be linked to physical losses and, ultimately, to financial and economic impacts.

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